

# Stock Market Volatility and International Business Cycle Dynamics: Evidence from OECD Economies

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## Abstract

How does a surprise movement in stock market volatility affect our forecasts of future output across countries? This paper studies the time series and cross-sectional responses of output to variation in stock market volatility across 27 countries over 40 years, controlling for a number of country-specific characteristics. High levels of stock market volatility are detrimental to future output growth not only after financial crises as previously emphasized in the literature, but also in non-crisis periods. Output growth and interest rates react negatively to a random shock to volatility and revert to their means quickly thereafter. Moreover, these results are robust after controlling for economic policy uncertainty, the level of financial development, and the direction of the market.

**Keywords:** Stock market volatility; Output growth; Heterogeneous panel causality.

**JEL Classification Numbers:** E4, E6, C23

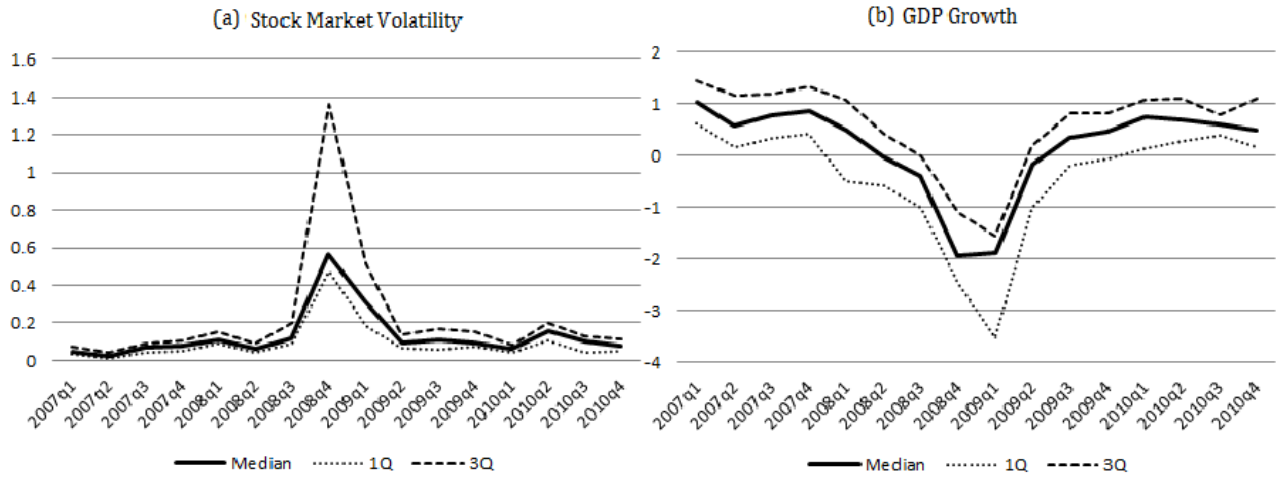
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## 1 Introduction

Figure 1 plots both the quarterly variance of the Morgan Stanley Capital International (MSCI) standard country equity index and real output growth. Two patterns are visible. First, output growth shrinks following spikes in stock market volatility. Second, the 2007 recession is undeniably among the most devastating since the Great Depression with prolonged output growth stagnancy and stark cross-country spillovers. How does a surprise movement in stock market volatility affect our forecasts of future output? I address this question using quarterly and monthly data across 27 countries from 1970 to 2012.

Figure 1: Stock Market Volatility and Output Growth (2007.Q1 - 2010.Q4)



**Note:** This figure presents the median, 1<sup>st</sup> quartile (denoted 1Q), and 3<sup>rd</sup> quartile (denoted 3Q) of stock market volatility and output growth across 27 countries from 2007.Q1 to 2010.Q4. Data are seasonally adjusted. The official NBER recession dates for the US are from 2007.Q2 to 2009.Q2, or 18 months.

While there are a plethora of studies linking increases in stock market volatility to declines in real output growth, they often abstract from incorporating cross-country dynamics. Given that many macroeconomic fundamentals are known to be driven by a variety of common world factors (see, for example, Gregory et al. (1997) and Crucini et al. (2011)), the analysis presented here thus fits in by accounting for these dynamics. Another source of novelty is the combined use of dynamic panel, panel vector autoregressive, and Granger-style panel causality analyses. These approaches are desirable because they allow for the analysis of the interaction of random shocks to real output growth and stock market volatility.

Before proceeding further, let us take a step back and review some of the most important works in the related literature. Using US data from 1834 to 1987, Schwert (1989) confirms that stock market volatility rose and interest rates fell on average during recessions in the pre-1987 period. As emphasized by Reinhart and Rogoff (2008), the aftermath of a financial crisis is typically associated with a large and persistent

decline in real output growth. This paper complements Reinhart and Rogoff (2008) by suggesting that future output growth reacts negatively and consistently to high stock market volatility in the previous period in both crisis *and* non-crisis periods. More recently, using a time-series VAR and multi-step Granger causality approach on Australian data, Rahman (2009) suggests that the volatility of the stock market tended to be associated with profound declines in real output growth and increases in both inflation and the unemployment rate. Similarly, Kanas and Ioannidis (2010) observe a strong relationship between stock market volatility and real activity in the UK over the period 1946-2002.

The research presented here also builds upon a growing literature that concerns the forecasting power of stock market volatility for future real economic activity, by incorporating international stock markets.<sup>1</sup> From a theoretical perspective, our results are supported by a strand of literature concerning the role of financial frictions in amplifying the response of the macroeconomic aggregates to various types of shocks. (e.g., Bernanke et al. (1999)). Most recently, Jermann and Quadrini (2012) attempt to explain such a mechanism using a DSGE framework in which firms cut back on hiring and investment when faced with stricter financial constraints. To preview the results, (1) a high level of stock market volatility corresponds to declines in real output growth during both crisis *and* non-crisis periods *regardless* of the direction of the market, (2) real output growth reacts negatively to a random shock to stock market volatility and reverts to its mean quickly thereafter, (3) all US recessions from 1970 to 2010 exhibit stark spill-over effects to other countries, and (4) economic policy uncertainty has a significant negative effect on output growth in the next period, but not to the extent that it undermines the effect of stock market volatility.

The paper is organized as follows. Section 2 estimates the effect of stock market volatility on output growth using a dynamic panel framework, controlling for a variety of sources of cross-country heterogeneity. Section 3, the focus of this paper, studies the interactions between output growth and stock market volatility in a panel VAR framework with impulse response analysis. Section 4 concludes.

## 2 Dynamic Panel Analysis

Table 1 presents descriptive statistics on stock market volatility and output growth across the 27 countries in our sample. Since past growth and stock market volatility may also contain certain characteristics of the state of the economy in the short run, lags of output growth and stock market volatility are included in the baseline specification in table 2. Stock market volatility is measured by the quarterly and monthly variance of the average daily MSCI country stock market index. With the exception of the MSCI index, all

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<sup>1</sup>See, for example, Campbell et al. (2011), Chauvet et al. (2012), and Fornari and Mele (2011).

data series are from the *OECD Stats* database.<sup>2</sup>

## 2.1 Estimation

Carlstrom and Fuerst (2003) characterize a reduced form of the Taylor rule in which the Federal Reserve changes interest rates in response to deviations of outputs growth and inflation from their expected growth paths. Following the literature, here we control for short-term interest rates as a proxy for monetary policy; two lags are included to allow for delays in implementation and effectiveness of monetary policy. Dummies that correspond to official US recessions from 1970 to 2010 are also included. Intuitively, these dummies account for spillovers from US recessions to other countries in our sample.<sup>3</sup>

Specifications (1)-(3) from table 2 present the baseline specifications with output as dependent variable and stock market volatility and interest rates as independent variables. Two measures of output are considered: the growth rate of industrial production and real GDP growth. Including industrial production as an alternative measure of output strengthens the results presented here as it allows our analysis to be carried on at the monthly frequency as well as the quarterly frequency.

While our measure of stock market volatility can capture changes in uncertainty in one country's stock market, valid criticism is that it ignores the direction of recent movements of stock prices and returns. Here we address this problem by including the log of the level of the MSCI index (at both monthly and quarterly frequencies) to account for the direction of the market.<sup>4</sup> Specifications (4)-(6) from table 2 present the results with the level of MSCI index. Since countries with better developed financial sectors tend to have higher and more sustained output growth, as widely supported in the literature, it is important to control for country-specific levels of financial development. To measure financial development, this paper follows Rousseau (2003) in using the ratio of broad money to real output. Table 2 presents the results with financial development in specifications (7)-(9).

Changes in economic policy also matter. Consider, for example, how firms have stronger incentives to invest in countries where corporate tax rates are comparably lower. Here we employ two proxies for changes in economic policy. Specifications (7)-(9) from table 2 control for a measure of overall tax levels, i.e., the ratio of tax revenue as a percentage of a country's real output. Table 3 (see, specifications (3)-(5))

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<sup>2</sup>Data are seasonally adjusted using the standard X-12-ARIMA procedure, unless otherwise specified.

<sup>3</sup>Official NBER business cycle dates, excluding the one from December 1969(IV) to November 1970 (IV). Another exception is the 2007 crisis, which is dated six months before the official NBER beginning date and continues until the end of 2010 (According the NBER, the peak was in 4<sup>th</sup> quarter 2007 technically, the recession lasted from January 2008 until June 2009, or 18 months). Recession dates are at <http://www.nber.org/cycles.html>, last accessed March 23, 2012.

<sup>4</sup>An alternative to MSCI index as a measure of market direction is market returns. The dynamic panel regression results with this measure is presented in table 3 in the technical appendix. One key result is that our conclusions remain qualitatively and quantitatively consistent regardless of which measure of market direction is employed.

extends the baseline results in table 2 to include a proxy for economic policy uncertainty index developed by Baker et al. (2013). Since the U.S. is the biggest economy in terms of real outputs in the sample, our results might be dominated by U.S. dynamics. This is, however, simply not the case. The results presented in Table 2 are consistent in terms of magnitude and sign without U.S. data in the sample. Given the nature of the time-series and cross-section dimensions of the dataset, all specifications are estimated using the Arellano-Bond GMM procedure with the first and second differences as instruments. Country-specific fixed effects are removed in the process of differencing as they do not vary over time.

## 2.2 Discussion

A key novel result from table 2 is that the first lag of stock market volatility has strong predictive power for output growth, even when controlling for country-specific effects. This result supports the idea that financial crises can negatively impact output growth in the next one or two quarters. On the one hand, stock market volatility only has a very short-term effect on growth as lags beyond one quarter do not show any significance, even at the 15% level. On the other hand, removing our measure of stock market volatility from the baseline specification in table 2 reduces our forecasting power approximately 5 percent.<sup>5</sup>

One common pattern across all specifications is that stock market volatility consistently has a *negative* effect on output growth, *regardless* of market direction, data frequency, and measure of output growth. In particular, whether the market is bullish or bearish does matter (as seen with the significance of the respective coefficients), but not to the extent that it takes away all the effects of our measure of uncertainty, i.e., stock market volatility. Quantitatively (see, table 2, specification (1)), a one percent increase in our volatility measure is expected to produce a 10.61 point decrease in industrial production index. High past interest rates correspond to declines in future output growth. In particular, a one percent increase in interest rate is expected to decrease output growth rate by approximately 2.15 percent with a two-month lag. This result is consistent with the perception that monetary policy changes are delayed in their effects on the economy. However, since each country has a different monetary and fiscal system, it is difficult to determine whether such a decrease is indeed attributable to the actions of the monetary authority.

The continued significance of the U.S. recession dummies across all specifications considered suggests

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<sup>5</sup>There is no consensus in the literature as to how  $R^2$  should be reported for the Arellano and Bond dynamic panel procedure due to complication in the dynamics of cross-sectional heterogeneity. On the other hand, as noted by Judson and Owen (2011), when the number of time-series dimension is large, the Arellano and Bond (1991) procedure can produce results that are equivalent to those of a fixed effects panel regression, which is reported in table 7. Thus, the values of  $R^2$  obtained from the fixed effects panel regression can be interpreted as a rough estimation of  $R^2$  for the Arellano and Bond procedure. Under this interpretation, moving from the specification without to the one with our measure of stock market volatility (from specification 1 to specification 2) increases  $R^2$  by approximately 5 percent.

that these recessions are not unique to the U.S. economy. This result is not surprising, given conclusions in the literature on world business cycles that macroeconomic fundamentals tend to share a common driving factor across nations.<sup>6</sup> Turning to Table 2 with an additional control for changes in tax policy overtime, the negative coefficients on taxes suggest a strong negative effect of changes in the tax rate on output growth. Quantitatively, consider, for example, the case with industrial production as the dependent variable, a one percent increase in tax rate corresponds to roughly a 0.174 percent decrease in real output growth.

Since a potential short-coming of the dynamic panel analysis is that causality can run in both directions, we next consider a three-step approach inspired by Hurlin (2005) to test potential causal relationships between market volatility and output growth. The results are presented in Table 5.<sup>7</sup> The causality running from stock market volatility to output growth is positive and significant. The magnitudes of the adjusted test statistics suggest that the direction of stock market volatility affecting growth is much *stronger* than the other way around, meaning that there is a potentially causal relationship from stock market volatility to output growth running in some cross sections. This result dovetails nicely with the dynamic panel analysis, which suggests that stock market volatility strongly predicts future output growth.

### 3 Impulse Responses

So far we have established the robust effects of increases in stock market volatility on future output growth. A natural question then arises concerning the dynamic effects of a random shock to stock market volatility on output growth forecasts at different horizons. Here we address this question using the impulse responses generated from a panel VAR framework.

#### 3.1 Estimation

Assuming all variables of interest are endogenous, panel VAR builds upon and extends the traditional VAR approach to a panel setting that allows for individual unobserved heterogeneity (Holtz-Eakin et al. (1988)). As a standard precursor to estimating the panel VAR, panel unit root tests are performed and the null of a unit root is rejected for the series employed here. These test results are reported in the statistical appendix. The approach of this section is similar to that of Love and Zicchino (2006), which focuses on the impulse response analysis of financial development and investment dynamics. Mathematically, a  $p$ -order

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<sup>6</sup>See, for example, Gregory et al. (1997) and Crucini et al. (2011).

<sup>7</sup>The results and the corresponding technical details of the test are provided in the Appendix.

VAR model is specified as follows:

$$Y_{i,t} = \alpha + \sum_{j=1}^p \beta_j Y_{i,t-j} + f_i + d_{c,t} + \varepsilon_t \quad (1)$$

where  $Y_{i,t}$  represents the three-variable vector of stock market volatility, interest rates, and GDP growth at time  $t$  for country  $i$ . This specification controls for cross-country heterogeneity using a country-specific dummy  $d_{c,t}$  and assumes that the shocks to current growth and stock market volatility can propagate through the dependent variables, which will become part of the explanatory variable set in the next period.

Similar to Love and Zicchino (2006), Monte Carlo simulations are used to calculate standard errors and generate confidence intervals.<sup>8</sup> Variance decompositions are reported over the course of twenty quarters, which covers the length of most recessions.<sup>9</sup> The panel VAR procedure requires that the underlying structure is homogeneous across each cross-section unit, which can be achieved by allowing for the fixed effects  $f_i$ . However, as emphasized by Love and Zicchino (2006), the fixed effects are often correlated with the regressors, which can be corrected by forward mean-differencing, which is referred to as the “Helmert” procedure Arellano and Bover (1995). Such a procedure allows for the use of lagged regressors as instruments to estimate the coefficients using GMM. Identification follows the literature in assuming long-run restrictions with lower triangular matrix and Cholesky decomposition.

### 3.2 Main Results

To maintain consistency with the dynamic panel analysis, here we also include output growth, interest rates, and stock market volatility with two lags. Two measures of output are employed as in Section 2. Industrial production is first-differenced, other variables are estimated in levels. Because of the lower-triangular matrix assumption, the order of the three-variable vector matters. Figure 2 present the impulse responses of industrial production to one-standard deviation shock to each of the three variables, with (a) stock market volatility ordered first, and (b) with stock market volatility ordered last. Figure 3 differs from Figure 2 in the sense that we replace industrial production with real GDP growth.

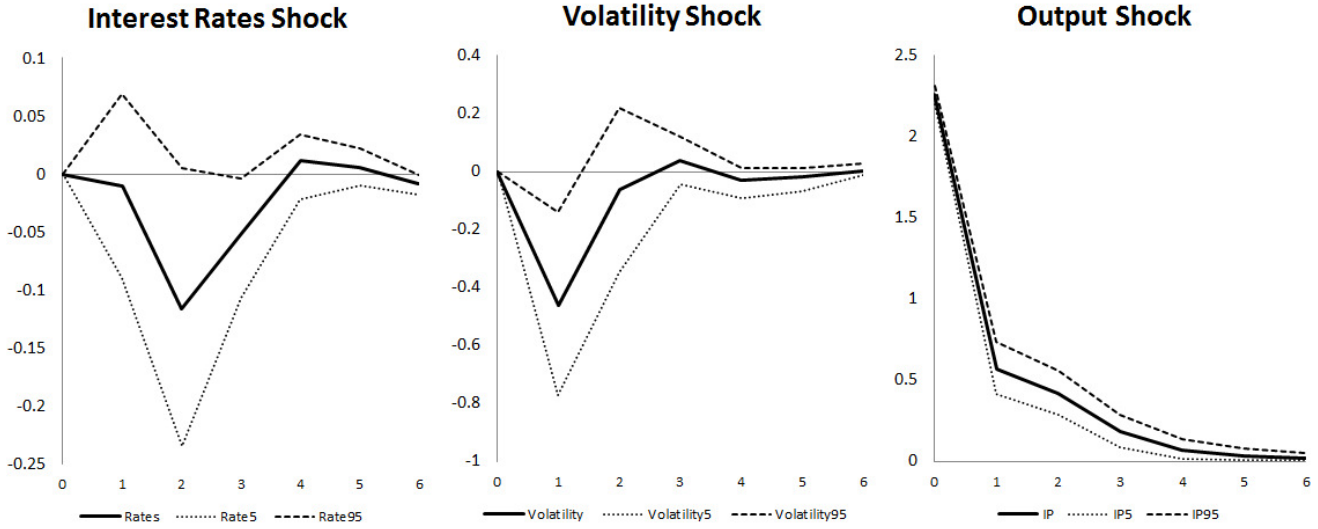
One common theme across our two measures of outputs is that innovations to market volatility have a significant dampening effect on output in the next quarter with output growth returning toward its

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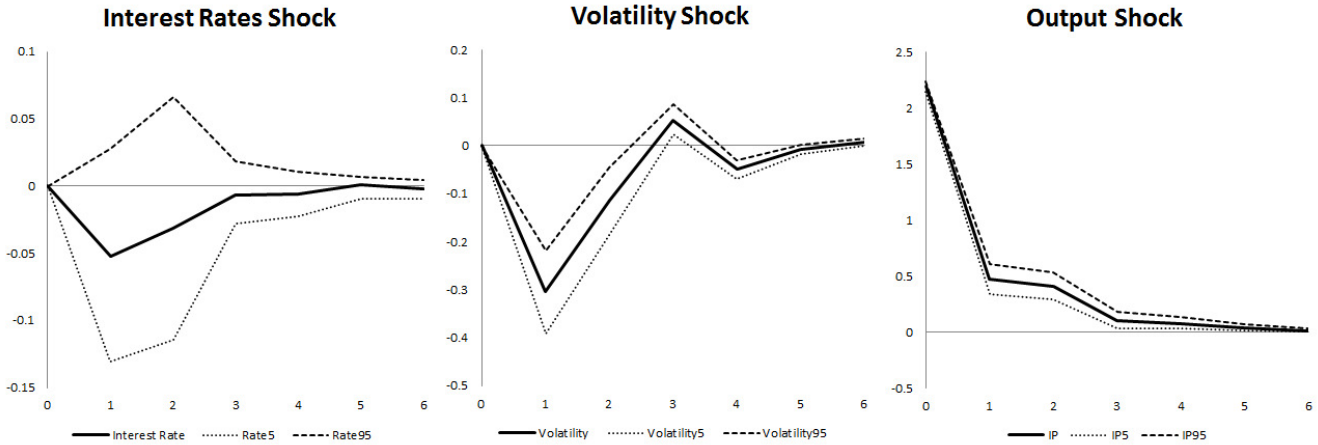
<sup>8</sup>To isolate the macro shocks and avoid taking into account the spillovers of shocks to impulse response function analysis requires the ability to decompose the residuals in a way that they become orthogonal. This section thus follows the convention by using Choleski decomposition to decompose the variance-covariance matrix of residuals Hamilton (1993). 2,000 replications are used for the Monte Carlo simulations to obtain the 5% error band for impulse response analysis.

<sup>9</sup>With the exception of the 2007 recession, past recessions often have a duration that is far shorter than five years.

(a) Volatility ordered first



(b) Volatility ordered last



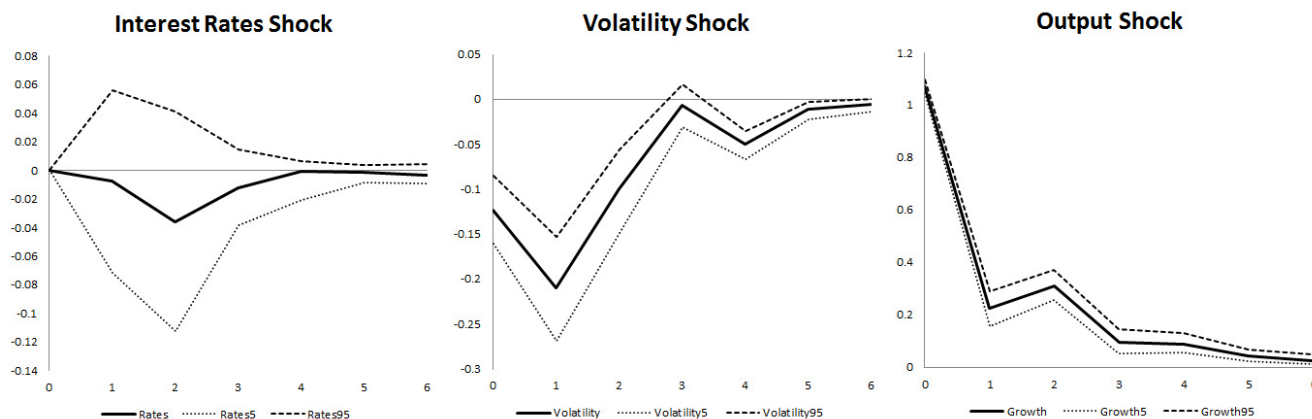
**Figure 2: Responses of Industrial Production**

**Note:** Impulse responses of quarterly industrial production growth to innovations of, from left to right, interest rates, stock market volatility, and industrial production (output) growth ( $p=2$ ). Lower triangular matrix is assumed for identification purpose. Order of the VAR system in the upper triplet: stock market volatility, interest rates, and output growth. Order of the VAR system in the lower triplet: interest rates, output growth, and stock market volatility. Dotted lines represent the 95% confidence bands.

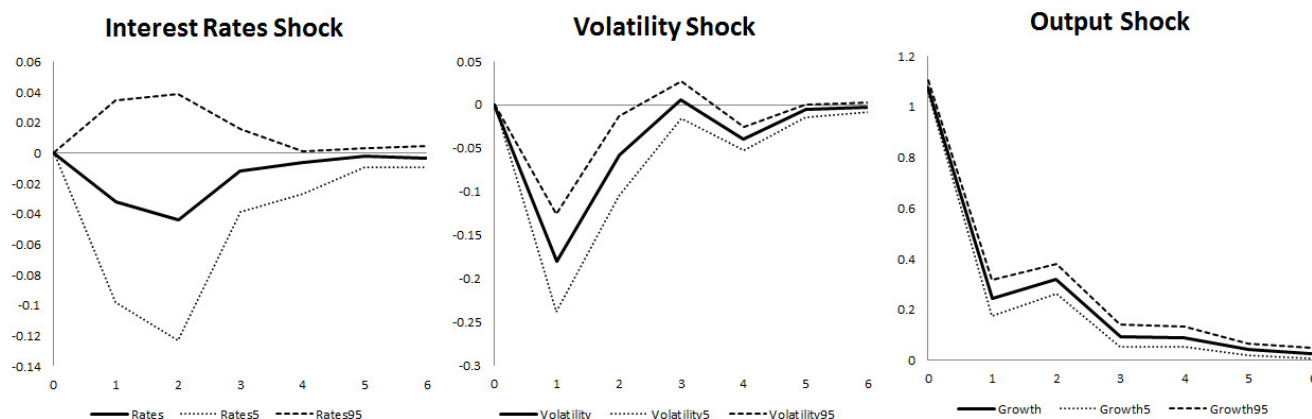
mean quickly thereafter. Turning to Figure 3 with real GDP growth, for example, a one-standard-deviation increase in the innovation to stock market volatility leads to a trough in output growth of 0.2 percent and a full growth recovery by the fourth quarter. Furthermore, since the responses of output to stock market volatility is significant at the 95% level in a short time window, stock market volatility contains important information about future output short horizons. Table 4 presents the fraction of the variance of the three variables accounted for by stock market volatility. Stock market volatility explains approximately



(a) Volatility ordered first



(b) Volatility ordered last



**Figure 3: Responses of Real GDP Growth**

**Note:** Impulse responses of quarterly real GDP growth to innovations of, from left to right, interest rates, stock market volatility, and real GDP (output) growth ( $p=2$ ). Lower triangular matrix is assumed for identification purpose. Order of the VAR system in the upper triplet: stock market volatility, interest rates, and output growth. Order of the VAR system in the lower triplet: interest rates, output growth, and stock market volatility. Dotted lines represent the 95% confidence bands.

2.7% of the total variance of growth while interest rates explain about 8.8% of the variation over a ten-quarter window. Past stock market volatility accounts for approximately 98% of variations in its future values; that is, information contained in past output and interest rates is almost exogenous to future market volatility. Given that country-specific effects are controlled for in the panel VAR system, this result is in fact, consistent with others in the literature on the determinants of international stock markets.<sup>10</sup>

<sup>10</sup>King and Wadhvani (1990) study the transmission of volatility between stock markets and suggest a strong co-movements among international stock markets despite “widely differing economic circumstances.” Along this dimension, King et al. (1994) analyze data from sixteen national stock markets and conclude that “only a small proportion of the time variation in the covariances between national stock markets can be accounted for by observable economic variables.”

One open question remains in terms of whether the innovations in stock market volatility are simply reflective of information already contained in real output and interest rates. The identification assumption of lower triangular matrix provides the perfect tool to address this question. In particular, here stock market volatility is ordered in a way such that it is orthogonal with respect to real output and responses of interest rates. In this sense, stock market volatility is the most exogenous among the three variables. Qualitatively, re-ordering of stock market volatility has virtually no impact on the results. Quantitatively, the impact of an innovation to stock market volatility is much higher on output when industrial production is used as a measure for real output and remains the same otherwise. The fact that the qualitative results presented here hold under both orthogonalizations dovetails nicely with the results from the Granger causality analysis that the direction of stock market volatility affecting output growth is much *stronger* than the other way around.

### 3.3 Extensions: Political Uncertainty, Oil Prices Volatility and Other Leading Index

So far we have established that past movements of stock market volatility contain significant information about next period output growth. In particular, output growth decreases when there is news about an increase in current stock market volatility. While stock market itself contains a wide range of information on economic uncertainty, one natural question remains to what degree such information contained in our measure of market volatility simply reflects that of other indicators of uncertainty.<sup>11</sup> This section attempts to answer this question by incorporating three measures of macroeconomic uncertainty into the panel VAR system as presented in section 3.2.<sup>12</sup>

The first measure, the leading indicator index from the OECD, is constructed from a wide range of macroeconomic indicators, one of which is the industrial production index. Since this measure combines information on macroeconomic events from a variety of sources, its behavior provides the answer to the question as to whether stock market volatility can provide *additional* information about next period output, in excess of other, perhaps more conventional, measures of future level of output. Table 3 in the appendix presents the dynamic panel regressions that include this leading index from the OECD. The continued significance of the coefficients on stock market volatility suggests that our results are robust even with the presence of such an indicator. Figure 4a plots the impulse responses of output to a one standard deviation

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<sup>11</sup>The author thanks an anonymous referee for suggesting these robustness checks.

<sup>12</sup>Unless otherwise noted, all impulse responses are generated from a four-variable panel VAR system with output growth, stock market volatility, interest rates, and each of the three extra measures of economic uncertainty. The order of each corresponding Cholesky decomposition is presented under each figure. The 95% confidence bands are generated using Monte Carlo simulations with 2,000 repetitions.

increase in the innovation in stock market volatility, interest rates, and the OECD leading indicator. On the one hand, output growth responds negatively to an orthogonal shock to stock market volatility and reverts to its mean quickly thereafter, which is consistent to the main results presented in section 3.2. On the other hand, this result also suggests that our measure of cross-country stock market volatility can provide additional forecasting information, even in the presence of a major leading indicator, such as the OECD leading index.

The second measure is oil price volatility, defined as the variance of the daily prices of one barrel of crude oil over a month, or a quarter.<sup>13</sup> One natural question arises as to whether the our measure of stock market volatility simply contains the information provided by the volatility of oil prices. This is not necessarily the case since stock market volatility can provide significant *additional* information for the forecast of output growth, as evidenced in the table 3 and the impulse responses of output growth in figure 4b. One common feature between stock market and oil prices volatility is that the effects of a one-time shock to their innovations tend to be short-lived, peaking approximately two months after the shocks and returning to the corresponding trends shortly thereafter.

The third measure is the economic policy uncertainty indices developed by Baker et al. (2013), which are constructed using micro data from a variety of newspapers sources.<sup>14</sup> Here we consider two indices, one for the US and one that covers most major economies in the European Union. Table 3 presents the results of the dynamic panel regression with these two measures of economic policy uncertainty. Figure 4c and 4d plot the impulse responses of output growth to a one-standard-deviation shock to stock market volatility, interest rates, and these two measures of economic policy uncertainty. As in the cases of the first two measures, our main results for stock market volatility presented in section 2.2 and 3.2 are robust, even with the inclusion of these economic policy uncertainty indices. It is interesting to note that a one-standard-deviation increase in US and European Union economic policy uncertainty decreases output growth, peaking at two and three months following the shocks with slow reversion back to their long-run growth paths. On the other hand, the effect of a random shock to political uncertainty peaks in around the same time as does a positive shock to stock market volatility but reverts to its corresponding mean far more slowly.

Three insights arise from incorporating these measures into our baseline specification. First, stock market volatility tends to exhibit significant forecasting power, even after controlling for more direct measures of uncertainty, such as oil prices volatility and the Baker et al. (2013) economic policy uncertainty indices. This result strengthens and extends the second insight that a positive increase in the innovation to stock

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<sup>13</sup>Data on oil prices was collected from the OECD website. Last accessed: Dec. 14, 2013.

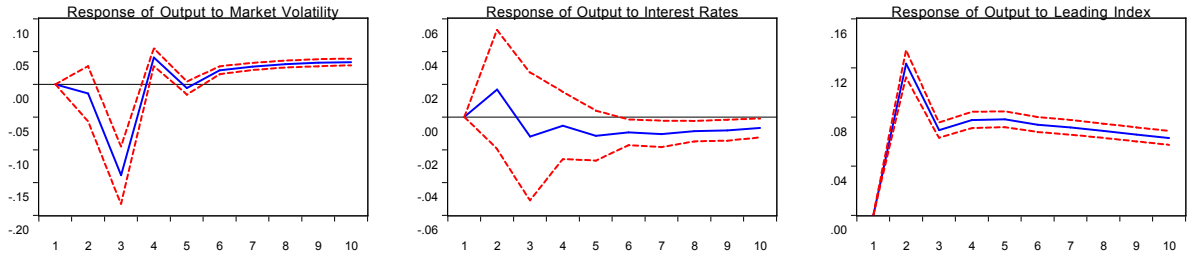
<sup>14</sup>See the appendix for a more elaborate description of the construction of the index.

market volatility dampens next period output growth, peaks after two months, and returns to its long-run trajectory quickly thereafter. Third, an increase in economic policy uncertainty and stock market volatility tends to have similar effects on the forecast of next period output growth in terms of timing of the peaks (i.e., around two to three months), but the effects of the former revert to their means far more slowly than do the effects of the latter.

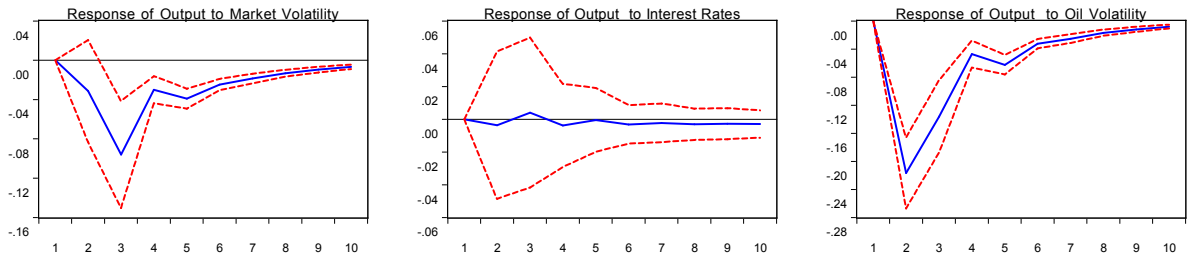
## **4 Conclusion**

This paper finds that movements in output growth can be predicted to innovations in stock market volatility in a sample of 27 countries, even when controlling for a number of cross-country characteristics. Stock market volatility appears to be a strong predictor of output growth in the succeeding one or two quarters; this result extends a growing literature on the forecasting power of stock market volatility on output growth by extending to the case of international stock markets. The analysis presented here also contributes the recent literature on financial crises and output growth by showing that output growth rate falls not only after a financial crises as previously emphasized by Reinhart and Rogoff (2008), but also in non-crisis periods with high stock market volatility. These results are robust after controlling for the levels of financial development and the direction of the market. It is also important to note that all US recessions from 1970 to 2010 exhibit strong spill-over effects to other countries, even when the US dynamics are removed. This result is consistent with the literature that documents strong co-movements of macroeconomic fundamentals across countries.

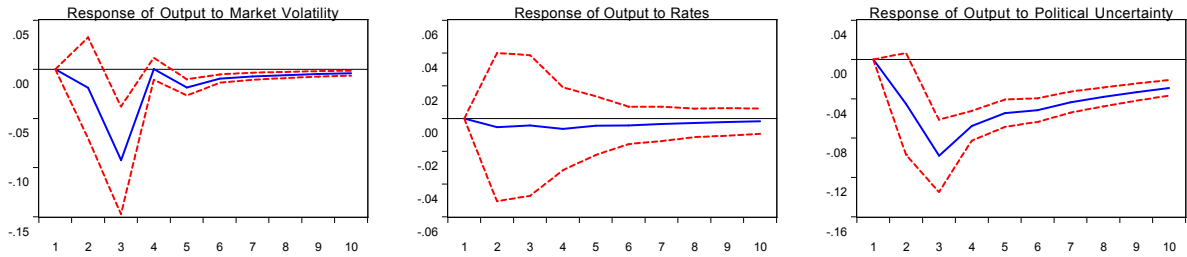
(a) Output, Stock Market Volatility, Interest Rates, and Leading Index



(b) Output, Stock Market Volatility, Interest Rates, and Oil Price Volatility



(c) Output, Stock Market Volatility, Interest Rates, and US Political Uncertainty



(d) Output, Stock Market Volatility, Interest Rates, and EU Political Uncertainty

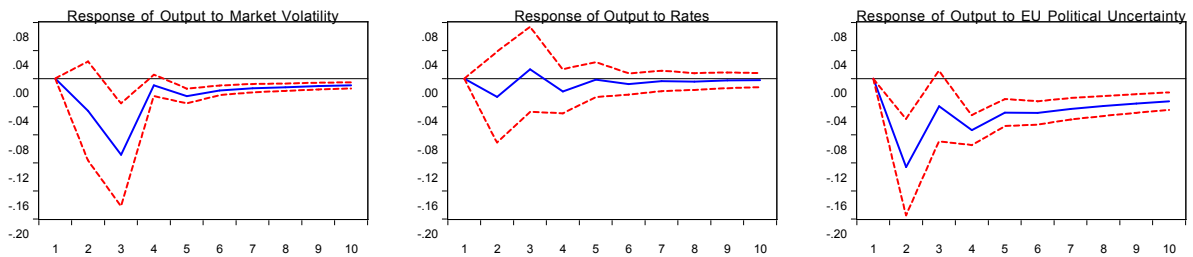


Figure 4: Responses of Output to Innovations of Uncertainty

**Note:** Monthly responses of industrial production in the 4-variable VAR system with industrial production, market volatility, interest rates, and an extra measure of volatility ( $p=2$ ). From up to down, that extra measure is OECD Leading Index, Oil Price Volatility, and the Baker et al. (2013) political uncertainty measure. Cholesky decomposition order is noted above the corresponding figure. Following section 3.2, the impulse responses are also re-ordered such that stock market volatility is ordered last. The results presented in this figure remain consistent. The dotted lines denote the 95 % confidence bands.

Table 1: Descriptive Statistics at Monthly Frequency

| Country      | Output      |             | Stock Market Volatility |              | Obs.         |
|--------------|-------------|-------------|-------------------------|--------------|--------------|
|              | Mean        | Std. Dev.   | Mean                    | Std. Dev.    |              |
| Austria      | 0.26        | 1.89        | 0.025                   | 0.019        | 273          |
| Belgium      | 0.16        | 2.14        | 0.018                   | 0.015        | 506          |
| Canada       | 0.16        | 1.10        | 0.013                   | 0.010        | 506          |
| Czech        | 0.21        | 2.73        | 0.029                   | 0.017        | 206          |
| Denmark      | 0.05        | 3.18        | 0.021                   | 0.012        | 302          |
| Estonia      | 0.03        | 2.30        | 0.033                   | 0.023        | 117          |
| Finland      | 0.19        | 2.50        | 0.031                   | 0.018        | 302          |
| France       | 0.07        | 1.36        | 0.020                   | 0.013        | 506          |
| Germany      | 0.12        | 1.71        | 0.019                   | 0.013        | 506          |
| Greece       | -0.29       | 2.91        | 0.033                   | 0.023        | 134          |
| Hungary      | 0.41        | 2.68        | 0.037                   | 0.026        | 206          |
| Iceland      | 0.90        | 4.75        | 0.078                   | 0.234        | 62           |
| Ireland      | 0.46        | 4.98        | 0.023                   | 0.016        | 290          |
| Italy        | 0.03        | 1.69        | 0.025                   | 0.014        | 401          |
| Japan        | 0.03        | 2.78        | 0.020                   | 0.010        | 119          |
| Korea        | 0.57        | 2.30        | 0.035                   | 0.023        | 254          |
| Luxembourg   | 0.36        | 3.33        | 0.043                   | 0.028        | 39           |
| Netherlands  | 0.12        | 2.80        | 0.020                   | 0.013        | 314          |
| Norway       | 0.08        | 4.50        | 0.027                   | 0.017        | 398          |
| Poland       | 0.53        | 2.32        | 0.040                   | 0.028        | 230          |
| Portugal     | 0.00        | 2.65        | 0.022                   | 0.013        | 290          |
| Slovak       | 0.36        | 3.31        | 0.029                   | 0.018        | 200          |
| Spain        | 0.04        | 1.89        | 0.023                   | 0.014        | 422          |
| Sweden       | 0.17        | 1.85        | 0.026                   | 0.015        | 362          |
| Turkey       | 0.13        | 5.28        | 0.052                   | 0.030        | 231          |
| UK           | 0.03        | 1.23        | 0.019                   | 0.011        | 410          |
| USA          | 0.18        | 0.76        | 0.014                   | 0.010        | 506          |
| <b>Total</b> | <b>0.17</b> | <b>2.60</b> | <b>0.025</b>            | <b>0.028</b> | <b>8,209</b> |

**Note:** Stock market volatility is the log of the level of the respective variables, seasonally adjusted. Output is the real GDP growth rate compared to the previous month, obtained from the OECD stats, seasonally adjusted. Market volatility is the variance of the MSCI index over the period of a month.

Table 2: Stock Market Volatility and Outputs

| Specifications |          | (1)                         | (2)                         | (3)                          | (4)                         | (5)                         | (6)                          | (7)                         | (8)                         | (9)                         |
|----------------|----------|-----------------------------|-----------------------------|------------------------------|-----------------------------|-----------------------------|------------------------------|-----------------------------|-----------------------------|-----------------------------|
| Lag            |          | IP                          |                             | GDP                          |                             | IP                          |                              | GDP                         |                             | GDP                         |
|                |          | Monthly                     | Quarterly                   | Quarterly                    | Monthly                     | Quarterly                   | Monthly                      | Quarterly                   | Quarterly                   |                             |
| Dep. Var       | 1        | -0.449***<br>[0.0118]       | 0.138***<br>[0.0244]        | -0.0121<br>[0.0257]          | -0.448***<br>[0.0119]       | 0.0827***<br>[0.0216]       | -0.0134<br>[0.0254]          | -0.244***<br>[0.0371]       | 0.100**<br>[0.0373]         | 0.0672<br>[0.0407]          |
|                | 2        | -0.149***<br>[0.0118]       | -0.00334<br>[0.0240]        | 0.0296<br>[0.0245]           | -0.149***<br>[0.0118]       | 0.012<br>[0.0212]           | 0.0228<br>[0.0243]           | -0.0565<br>[0.0352]         | 0.0209<br>[0.0374]          | 0.0971*<br>[0.0389]         |
| Volatility     | 1        | <b>-10.61***</b><br>[1.850] | <b>-44.91***</b><br>[7.250] | <b>-27.25***</b><br>[3.254]  | <b>-9.966***</b><br>[1.887] | <b>-33.57***</b><br>[6.383] | <b>-24.38***</b><br>[3.265]  | <b>-13.54***</b><br>[2.313] | <b>-48.36***</b><br>[9.873] | <b>-28.13***</b><br>[4.676] |
|                | 2        | -16.00***<br>[1.873]        | -5.267<br>[7.382]           | -7.871*<br>[3.370]           | -15.38***<br>[1.891]        | 2.945<br>[6.365]            | -5.784<br>[3.361]            | -15.11***<br>[2.324]        | 10.62<br>[10.22]            | -1.672<br>[4.943]           |
| Rates          | 1        | -0.0135**<br>[0.0048]       | 0.00316<br>[0.0247]         | 0.00279<br>[0.0112]          | -0.0152**<br>[0.00578]      | -0.0107<br>[0.012]          | -0.00201<br>[0.0112]         | -0.00989<br>[0.00515]       | 0.000471<br>[0.0258]        | 0.00958<br>[0.0122]         |
|                | 2        | -0.0215***<br>[0.0048]      | -0.0326<br>[0.0229]         | -0.0036<br>[0.010]           | -0.0167**<br>[0.0058]       | -0.00913<br>[0.011]         | -0.00851<br>[0.0103]         | -0.00601<br>[0.00519]       | -0.028<br>[0.0242]          | -0.00211<br>[0.0114]        |
| Finan. Acco.   | 1        |                             |                             |                              |                             |                             |                              | <b>20.47***</b><br>[3.084]  | -3.382<br>[4.306]           | 2.158<br>[2.170]            |
|                | 2        |                             |                             |                              |                             |                             |                              | <b>10.10***</b><br>[2.924]  | 7.862<br>[4.221]            | <b>7.340***</b><br>[2.105]  |
| Market Level   |          |                             |                             |                              | <b>0.783*</b><br>[0.305]    | <b>2.122***</b><br>[0.273]  | <b>0.756***</b><br>[0.109]   | -0.112<br>[0.194]           | 0.381<br>[0.222]            | 0.0854<br>[0.0982]          |
|                | Tax      |                             |                             |                              |                             |                             |                              | -0.174**<br>[0.0624]        | -0.123***<br>[0.0287]       |                             |
| Recession      |          | <b>-0.431**</b><br>[0.156]  | <b>-1.364***</b><br>[0.184] | <b>-0.775***</b><br>[0.0820] | <b>-0.370*</b><br>[0.158]   | <b>-1.315***</b><br>[0.161] | <b>-0.777***</b><br>[0.0813] | <b>-0.434**</b><br>[0.155]  | <b>-1.678***</b><br>[0.248] | <b>-0.922***</b><br>[0.115] |
|                | Constant | 0.950***<br>[0.0737]        | 1.143***<br>[0.129]         | 1.093***<br>[0.0694]         | -3.599<br>[1.878]           | -11.97***<br>[1.686]        | -3.693***<br>[0.694]         | 1.765<br>[1.258]            | -1.307<br>[1.454]           | 0.454<br>[0.654]            |
| Obs.           |          | 8,039                       | 2,618                       | 2,142                        | 8,039                       | 2,618                       | 2,142                        | 3,545                       | 1,023                       | 1,023                       |

Standard errors in brackets

\* p<0.05, \*\* p<0.01, \*\*\* p<0.001

Table 3: Regression Results using Monthly Data

| Specifications               |     | (1)                  | (2)                  | (3)                   | (4)                   | (5)                   |
|------------------------------|-----|----------------------|----------------------|-----------------------|-----------------------|-----------------------|
|                              | Lag | Output               | Output               | Output                | Output                | Output                |
| <b>Output</b>                | 1   | 0.458***<br>(0.011)  | 0.458***<br>(0.011)  | 0.425***<br>(0.012)   | 0.386***<br>(0.015)   | 0.381***<br>(0.015)   |
|                              | 2   | 0.369***<br>(0.011)  | 0.369***<br>(0.011)  | 0.336***<br>(0.012)   | 0.314***<br>(0.015)   | 0.319***<br>(0.015)   |
| <b>Leading Indicator</b>     | 1   | 1.780***<br>(0.133)  | 1.832***<br>(0.142)  | 1.232***<br>(0.158)   | 0.737***<br>(0.195)   | 0.553**<br>(0.199)    |
|                              | 2   | -1.637***<br>(0.135) | -1.689***<br>(0.143) | -1.027***<br>(0.161)  | -0.505*<br>(0.199)    | -0.327<br>(0.203)     |
| <b>Rates</b>                 | 1   | 0.00640<br>(0.005)   | 0.00640<br>(0.005)   | 0.00909<br>(0.005)    | 0.0267**<br>(0.009)   | 0.0238**<br>(0.009)   |
|                              | 2   | -0.00311<br>(0.005)  | -0.00301<br>(0.005)  | 0.00255<br>(0.005)    | 0.0294**<br>(0.009)   | 0.0288**<br>(0.009)   |
| <b>Volatility</b>            | 1   | -10.61***<br>(1.640) | -10.65***<br>(1.663) | -5.354**<br>(1.869)   | -6.467**<br>(2.395)   | -4.300<br>(2.441)     |
|                              | 2   | -21.56***<br>(1.622) | -21.24***<br>(1.627) | -17.80***<br>(1.791)  | -19.18***<br>(2.310)  | -17.88***<br>(2.329)  |
| <b>Recession</b>             |     | 0.0241<br>(0.060)    | 0.0215<br>(0.060)    | 0.141*<br>(0.071)     | -0.00534<br>(0.078)   | 0.0652<br>(0.080)     |
|                              |     |                      |                      |                       |                       |                       |
| <b>Market Return</b>         | 1   |                      | -17.98*<br>(7.147)   | -30.40***<br>(7.902)  | -33.99***<br>(10.106) | -33.23**<br>(10.119)  |
|                              | 2   |                      | 8.571<br>(7.250)     | -4.515<br>(8.021)     | -1.397<br>(10.327)    | -3.046<br>(10.345)    |
| <b>Political Uncertainty</b> | USA |                      |                      | -0.0147***<br>(0.001) | -0.00407*<br>(0.002)  | -0.00474*<br>(0.002)  |
|                              | EUR |                      |                      |                       | -0.0184***<br>(0.002) | -0.0167***<br>(0.002) |
| <b>Oil Price Volatility</b>  |     |                      |                      |                       |                       | -8.744***<br>(1.839)  |
| <b>Constant</b>              |     | 4.314***<br>(0.327)  | 4.306***<br>(0.327)  | 6.243***<br>(0.394)   | 10.43***<br>(0.652)   | 11.11***<br>(0.668)   |
|                              |     |                      |                      |                       |                       |                       |
| <b>Obs.</b>                  |     | 7,644                | 7,644                | 6,391                 | 4,059                 | 4,059                 |

Standard errors in brackets

\* p<0.05, \*\* p<0.01, \*\*\* p<0.001



**Table 4: Variance Decompositions for 20 quarters**

|                   | Past Periods | Growth  | Market volatility | Interest Rate |
|-------------------|--------------|---------|-------------------|---------------|
| Output Growth     | 10           | 0.97281 | 0.02702           | 0.00017       |
| Market volatility | 10           | 0.00132 | 0.99858           | 0.00009       |
| Interest Rate     | 10           | 0.08833 | 0.00805           | 0.90361       |
| Output Growth     | 20           | 0.97268 | 0.02702           | 0.00031       |
| Market volatility | 20           | 0.00133 | 0.99850           | 0.00017       |
| Interest Rate     | 20           | 0.09348 | 0.00723           | 0.89929       |

**Table 5: Panel Causality Test**

| Lags                                     | Z     |
|------------------------------------------|-------|
| Market volatility weakens growth         |       |
| 1                                        | 38.18 |
| 2                                        | 12.49 |
| 3                                        | 5.58  |
| Growth increases stock market volatility |       |
| 1                                        | 12.16 |
| 2                                        | 5.85  |
| 3                                        | 4.47  |

**Note:** The adjusted Z-statistics follows the  $N(0,1)$  standard normal distribution.

## Appendix

### A Data

The dataset covers 27 OECD countries across 40 years on a quarterly and monthly bases, seasonally adjusted. With the exception of stock market volatility, all data are from the *OECD Stats* database. The market data index comes from the Morgan Stanley Capital International (MSCI) standard country equity index that covers the majority of the country's stocks. The precise definition of the index is as follows:

To construct a country index, every listed security in the market is identified. Securities are free float adjusted, classified in accordance with the Global Industry Classification Standard (GICS®), and screened by size, liquidity, and minimum free float.<sup>15</sup>

Market data are collected and reported daily. I obtain stock market volatility by taking the variance of the logarithm of the country specific market index over the course of one or three months. Interest rates are the short-term interest rate measure by the OECD. Taxes are the ratios of tax revenues over real outputs,

<sup>15</sup>This definition is from <http://www.msci.com/products/indices/tools/> and were accessed on April 12, 2012. All market data were collected using Thomson Reuters Datastream.

which covers both corporate tax and personal income tax. Baker et al. (2013) construct a number of policy uncertainty indices using micro-data from a large number of newspaper articles. Two series are employed here: one for the US and one common index for the Euro zone. In terms of methodology, the index is constructed from three components, a description of which can be summarized as follows:<sup>16</sup>

One component quantifies newspaper coverage of policy-related economic uncertainty. A second component reflects the number of federal tax code provisions set to expire in future years. The third component uses disagreement among economic forecasters as a proxy for uncertainty.

Here we calculate a measure of the market return by subtracting the log of MSCI index from the log of the same index in the succeeding period. The result that uses this measure of market returns is presented in Table 3.

## B Panel Units Roots Test

To test for unit roots in heterogeneous panels, two procedures are often used: (1) Levin et al. (2002), LL hereafter and (2) Im et al. (2003), IPS hereafter. Under LL specification, the error terms are independent and the persistence parameter is the same across sections, which requires the presence of a strongly balanced panel. As in the literature, the tests are based on the following model:

$$\Delta y_{i,t} = \alpha_i + \beta_i y_{i,t-1} + \delta_i t + \sum_{p=1}^{P_i} \theta_i^{(p)} \Delta y_{i,t-p} + \varepsilon_{i,t} \quad (2)$$

where  $y_{i,t}$  denotes the variable  $y$  observed for the  $i^{th}$  cross-sectional unit and time  $t$  unit.

Instead of assuming a common unit root process, the IPS procedure tests for individual unit roots processes, which means that the IPS test is more suitable for a less balanced dataset. Given the highly unbalanced nature of our dataset, this paper uses the IPS test; null hypothesis is that all panels contain unit roots, against the alternative hypothesis that some of the panel contains unit roots. By rejecting the hypothesis that all panels contain unit roots (as in table 6), the test also satisfies the first requirement in the three-step panel Granger causality procedure in the next section.

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<sup>16</sup>The except was retrieved from <http://www.policyuncertainty.com/methodology.html> on Dec. 20, 2013. All corresponding data on the policy uncertainty index was also collected from the same website.

**Table 6: IPS panel unit root test**

| Variable                | Test statistics |
|-------------------------|-----------------|
| Stock Market Volatility | -21.2078        |
| Output Growth (GDP)     | -21.4656        |
| Tax Level               | -2.5173         |
| Financial Accommodation | -3.9314         |
| Interest Rate           | -4.7784         |

**Note:** Significant values of the test statistics reject the null hypothesis. This specification does include time trend. Ho: All panels contain unit roots. Ha: Some panels are stationary.

## C Panel Causality Test

I test for potential causal relationship between market volatility and output growth using the specification by Hurlin and Venet (2001). Like the traditional Granger causality test, this procedure requires first and foremost the assumption that the variables under investigation are covariance-stationary. As a result, I adopt a three-step approach as follows:

- First, test for stationarity in the testing variables. Note that this is a requirement for the panel causality test to be valid.
- Second, obtain the individual Wald test for each panel by testing the hypothesis described in equation (6).
- Lastly, obtain the adjusted test statistics  $Z_{NL}$  for the *unbalanced* panel dataset that converges in distribution to the standard normal as shown in Hurlin (2005).

Upon verifying the stationarity of the growth and market volatility, the causality relationship between those two variables is tested by following Hurlin (2005), which assumes the following specification:

$$y_{i,t} = \alpha_i + \sum_{p=1}^P \gamma_i^{(p)} y_{i,t-k} + \sum_{p=1}^P \beta_i^{(p)} x_{i,t-k} + \varepsilon_{i,t} \quad (3)$$

where  $P$  denotes the number of maximum number of lags specified. The null hypothesis to be tested is:

$$H_o : \beta_i^{(1)} = \beta_i^{(2)} = \dots = \beta_i^{(P)} = 0, \forall i = 1, \dots, N \quad (4)$$

This *Homogenous Non-Causality* (HNC) hypothesis, allows for causality from  $x$  to  $y$  for some but no all individuals. This approach consists of two steps: (1) compute each individual  $i$  Wald statistics under the

null, and (2) compute the average of the Wald statistics from the first step. Since the dataset is unbalanced, it is important to adjust the previously-obtained average statistics to account for this heterogeneity across the cross section dimension. Following Hurlin and Venet (2001), I adjust the average Wald statistics using the following formula<sup>17</sup>:

$$Z_{N,T} = \sqrt{N} \left( W - \frac{P}{N} \sum_{i=1}^N \frac{T_i - 2P - 1}{T_i - 2P - 3} \right) \times \left( \frac{2P}{N} \sum_{i=1}^N \frac{(T_i - 2P - 1)^2 \times (T_i - P - 3)}{(T_i - 2P - 3)^2 \times (T_i - 2P - 5)} \right) \quad (5)$$

where the adjusted  $Z_{N,T}$  follows a standard normal distribution as shown in Hurlin (2005). A statistically significant value of  $Z_{N,T}$  implies that we can reject the null hypothesis that there is no causality running in the direction of testing.

## D Additional Tables and Figures

This section includes extra tables that have been previously mentioned in the main text of the paper. In particular, Tables 3 and 7 present the regression results under dynamic panels and fixed effects least squares dummies regressions, respectively. In terms of notation, the prefix L. and L2. denote the first and second lag, respectively.

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<sup>17</sup>One important technical requirement of the adjustment is that  $T_i > 5 + 2P$ , which are satisfied across the dataset.

Table 7: Regression Results using Monthly Data (Fixed Effects)

| Specifications               |     | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   |
|------------------------------|-----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                              | Lag | Output                | Output                | Output                | Output                | Output                |
| <b>Output</b>                | 1   | -0.362***<br>(-32.20) | -0.367***<br>(-32.70) | -0.439***<br>(-38.74) | -0.439***<br>(-38.73) | -0.434***<br>(-35.04) |
|                              | 2   | -0.0830***<br>(-7.38) | -0.0870***<br>(-7.75) | -0.147***<br>(-12.98) | -0.147***<br>(-12.95) | -0.148***<br>(-11.96) |
| <b>Rates</b>                 | 1   | -0.00542<br>(-1.34)   | -0.00468<br>(-1.16)   | 0.0000571<br>(0.01)   | 0.000198<br>(0.05)    | -0.000758<br>(-0.19)  |
|                              | 2   | -0.00654<br>(-1.62)   | -0.00637<br>(-1.58)   | -0.00533<br>(-1.40)   | -0.00525<br>(-1.38)   | -0.00490<br>(-1.21)   |
| <b>Recession</b>             |     | -0.529***<br>(-9.58)  | -0.467***<br>(-8.40)  | -0.274***<br>(-5.03)  | -0.271***<br>(-4.98)  | -0.283***<br>(-4.43)  |
| <b>Volatility</b>            | 1   |                       | -2.801**<br>(-3.17)   | -2.230<br>(-1.48)     | -1.630<br>(-1.07)     | -1.516<br>(-0.87)     |
|                              | 2   |                       | -5.389***<br>(-6.08)  | -10.80***<br>(-7.13)  | -10.50***<br>(-6.92)  | -10.81***<br>(-6.35)  |
| <b>Leading Indicator</b>     | 1   |                       |                       | 0.411<br>(1.13)       | 0.355<br>(0.96)       | 0.497<br>(1.24)       |
|                              | 2   |                       |                       | 1.647***<br>(4.41)    | 1.632***<br>(4.36)    | 1.438***<br>(3.53)    |
| <b>Market Return</b>         | 1   |                       |                       |                       | -3.745<br>(-0.55)     | -2.176<br>(-0.29)     |
|                              | 2   |                       |                       |                       | 24.78***<br>(3.64)    | 25.11***<br>(3.31)    |
| <b>Political Uncertainty</b> | USA |                       |                       |                       |                       | -0.000251<br>(-0.29)  |
| <b>Constant</b>              |     | 0.407***<br>(11.17)   | 0.587***<br>(13.45)   | 0.320***<br>(5.41)    | 0.301***<br>(5.08)    | 0.361***<br>(3.44)    |
| <b>Obs.</b>                  |     | 7,897                 | 7,897                 | 7,897                 | 7,897                 | 6,401                 |

t statistics in parentheses

\* p<0.05, \*\* p<0.01, \*\*\* p<0.001

## References

- ARELLANO, M. AND S. BOND (1991): “Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations,” *Review of Economic Studies*, 58, 277–297.
- ARELLANO, M. AND O. BOVER (1995): “Another look at the instrumental variable estimation of error component models,” *Journal of Econometrics*, 68, 29–51.
- BAKER, S. R., N. BLOOM, AND S. J. DAVIS (2013): “Measuring Economic Policy Uncertainty,” *Working Paper*.
- BERNANKE, B., M. GERTLER, AND S. GILCHRIST (1999): “The Financial Accelerator in a Quantitative Business Cycle Framework,” *NBER Working Paper*.
- CAMPBELL, J. Y., M. LETTAU, B. G. MALKIEL, AND Y. XU (2011): “Have Individual Stocks Become More Volatile? An Empirical Exploration of Idiosyncratic Risk,” *Journal of Finance*, 56, 1–43.
- CARLSTROM, C. T. AND T. S. FUERST (2003): “The Taylor Rule: a guidepost for monetary policy,” *Federal Reserve Bank of Cleveland Economic Commentary*.
- CHAUVET, M., Z. SENYUZ, AND E. YOLDAS (2012): “What Does Realized Volatility Tell Us About Macroeconomic Fluctuations?” *Working paper, Federal Reserve Board*.
- CRUCINI, M., A. KOSE, AND C. OTROK (2011): “What are the driving forces of international business cycles?” *Review of Economic Dynamics*, 14, 156–175.
- FORNARI, F. AND A. MELE (2011): “Financial Volatility and Economic Activity,” *Working paper, London School of Economics*.
- GREGORY, A. W., A. C. HEAD, AND J. RAYNAULD (1997): “Measuring World Business Cycles,” *International Economic Review*, 38, 677–701.
- HAMILTON, J. D. (1993): *Time series analysis*, Princeton University Press.
- HOLTZ-EAKIN, D., W. NEWEY, AND H. S. ROSEN (1988): “Estimating Vector Autoregressions with Panel Data,” *Econometrica*.
- HURLIN, C. (2005): “Un test simple de l’hypothèse de non causalité dans un modèle de panel hétérogène,” *Revue Economique*.

- HURLIN, C. AND B. VENET (2001): “Granger Causality Tests in Panel Data Models with Fixed Coefficients,” *Working Paper Eurisco 2001-09, University of Paris Dauphine*.
- IM, K. S., M. H. PESARAN, AND Y. SHIN (2003): “Testing for unit roots in heterogeneous panels,” *Journal of Econometrics*, 115, 53–74.
- JERMANN, U. AND V. QUADRINI (2012): “Macroeconomic Effects of Financial Shocks,” *American Economic Reviews*, 102, 238–271.
- JUDSON, R. A. AND A. L. OWEN (2011): “Estimating dynamic panel data models: a guide for macroeconomists,” *Economic Letters*, 65, 9–15.
- KANAS, A. AND C. IOANNIDIS (2010): “Causality from Real Stock Returns to Real Activity: Evidence of Regime-Dependence,” *International Journal of Finance and Economics*, 15, 180–197.
- KING, M., E. SENTANA, AND S. WADHWANI (1994): “Volatility and Links Between National Stock Markets,” *Econometrica*, 62, 901–933.
- KING, M. A. AND S. WADHWANI (1990): “Transmission of Volatility Between Stock Markets,” *Review of Financial Studies*, 3, 5–33.
- LEVIN, A., C.-F. LIN, AND J. C.-S. CHU (2002): “Unit root tests in panel data: Asymptotic and finite sample properties,” *Journal of Econometrics*, 108, 1–24.
- LOVE, I. AND L. ZICCHINO (2006): “Financial Development and Dynamic Investment Behaviour,” *The Quarterly Review of Economics and Finance*, 46, 190–210.
- RAHMAN, M. A. (2009): “Industry-level stock returns volatility and aggregate economic activity in Australia,” *Applied Financial Economics*, 19, 509–525.
- REINHART, C. M. AND K. S. ROGOFF (2008): “Is the 2007 U.S. Sub-Prime Financial Crisis So Different? An International Historical Comparison,” *American Economics Review*, 82, 339–344.
- ROUSSEAU, P. B. (2003): “Historical Perspectives on Financial Development and Economic Growth,” *NBER Working Papers*.
- SCHWERT, W. G. (1989): “Business Cycles, Financial Crises, and Stock Volatility,” *Carnegie Rochester Conference Series on Public Policy*, 31, 83–125.